

Compiler Design and Construction

Lab Index

Lab 1: Implementing Lexical Analyzer

Implement a lexical analyzer to recognize identifiers, keywords, comments, strings, operators, and constants. Display token type and lexeme.

Lab 2: Implementing Symbol Table Operations

Implement a symbol table to demonstrate the operations: insert, lookup and display. Maintain attributes such as identifier name, type, and scope.

Lab 3: Recursive Descent Parser

Implement a recursive descent parser for the grammar

$$\begin{aligned} S &\rightarrow a A b \\ A &\rightarrow a \mid \varepsilon \end{aligned}$$

Lab 4: Implementation of Shift-Reduce Parsing.

Implement Shift-Reduce parsing for the following grammar and input string:
a +a*a.

$$E \rightarrow E + E$$

$$E \rightarrow E * E$$

$$E \rightarrow (E)$$

$$E \rightarrow a$$

Lab 5: Write a program to generate closure set on LR(0) items for the grammar:

$$S \rightarrow A B$$

$$A \rightarrow a$$

$$B \rightarrow b$$

Lab 6: Intermediate Code Generation

Write a program to generate **three-address code** for arithmetic assignment statement.

Lab 7: Target Code Generation

Write a program to generate **target code** for a simple register-based machine.